

$$\text{Dynamics: } \dot{x}(t) = -2x(t) + u(t)$$

$$\text{Constraints: } x(0) = 2$$

$$x(1) = 0$$

$$\text{cost functional: } J(u) = \int_0^1 u(t)^2 dt$$

$$f(x, u, t) = -2x(t) + u(t)$$

$$h(x_{tf}, t_f) = 0$$

$$g(x, u, t) = u(t)^2$$

$$H = g + pf$$

$$\begin{aligned} H(x, u, p) &= u^2 + p(-2x + u) \\ &= u^2 - 2px + pu \end{aligned}$$

$$\text{NOC: } \dot{x}^* = \frac{\partial H}{\partial p} = -2x^* + u^*$$

$$\dot{p}^* = -\frac{\partial H}{\partial x} = 2p^*$$

$$u^* = \arg\min_u H$$

$$\therefore 0 = \frac{\partial H}{\partial u} = 2u^* + p^*$$

$$\therefore u^* = -\frac{1}{2}p^*$$

$$\therefore \dot{p}^* = 2p^*$$

$$\therefore p^*(t) \text{ must be like } Ce^{2t}$$

$$\therefore u = -\frac{C}{2}e^{2t}$$

$$\therefore \dot{x} = -2x - \frac{C}{2}e^{2t}$$

$$\dot{x} + 2x = -\frac{C}{2}e^{2t}$$

$$\dot{x}e^{2t} + 2e^{2t}x = -\frac{C}{2}e^{4t}$$

$$\frac{d(e^{2t}x(t))}{dt} = -\frac{C}{2}e^{4t}$$

$$\therefore e^{2t}x(t) = -\frac{C}{8}e^{4t} + D$$

$$\therefore x(t) = -\frac{C}{8}e^{2t} + De^{-2t}$$

$$\therefore \begin{cases} x(0) = 2 \\ x(1) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{C}{8}e^0 + De^0 = 2 \\ -\frac{C}{8}e^2 + De^{-2} = 0 \end{cases}$$

$$\therefore C = 8D - 16$$

$$\Rightarrow \begin{cases} C = \frac{16}{e^4 - 1} \\ D = \frac{2e^4}{e^4 - 1} \end{cases}$$

$$\therefore x^*(t) = -\frac{2}{e^4 - 1}e^{2t} + \frac{2}{e^4 - 1}e^{4-2t}$$

$$\therefore u^*(t) = -\frac{8}{e^4 - 1}e^{2t}$$